



Deptora

**AI-Powered Administrative Assistant
for
Departmental Automation**



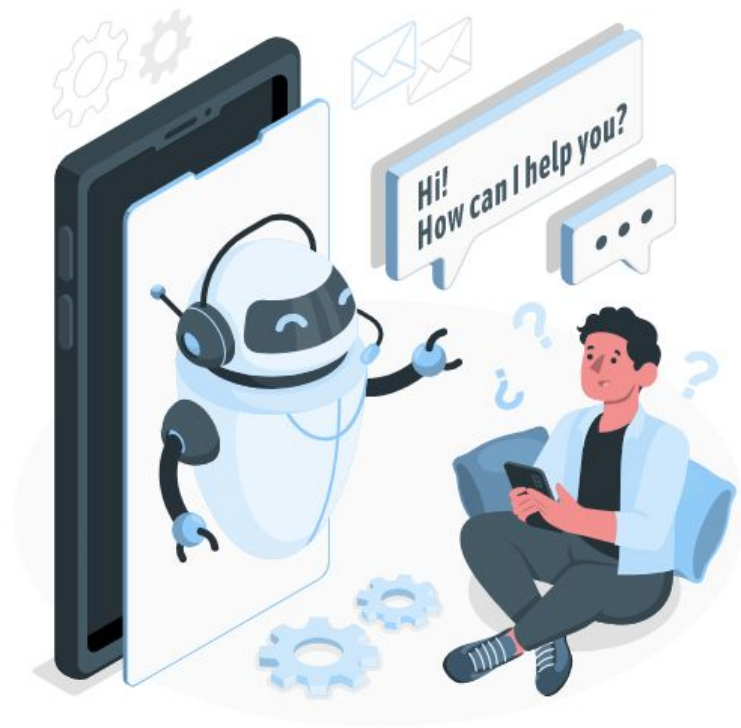
Capstone Project- 400C Presentation

Our Supervisor,
Md Al-Imran
Senior Lecturer
Department of Computer Science &
Engineering

Presented By,

1. **Swabirah Iffat Binte Ameer**
ID: 2022-3-60-067
2. **Samia Rahman**
ID: 2022-3-60-087
3. **Mossamod Kabira Kalam**
ID: 2022-3-60-079
4. **Ishrat Jahan Anika**
ID: 2022-3-60-195

Agenda



Introduction

Problem

Research Gap

Why Existing AI Isn't Enough

Project Aim

Objectives

Overall System

Document Ingestion

Query Processing

Hybrid Retrieval

Domain Restriction

Response Validation

Interface

Experimental Setup

Result

Limitations

Future Work

Conclusion



Introduction



THE CONTEXT

University departments handle a large number of repetitive student queries about academic policies, registration, deadlines, courses, forms, and procedures.



THE CHALLENGE

Traditional support methods are often manual, slow, static, and difficult to scale, while general-purpose AI may not provide the required privacy, accuracy, and domain control.



OUR APPROACH

DEPTORA uses local LLM + RAG + hybrid retrieval + domain restriction to provide grounded answers from official departmental documents.

STUDENT



Questions about courses, policies, deadlines, forms, procedures, etc.



DEPTORA



Domain Restriction



Hybrid Retrieval



Secure & Private



Local LLM



Grounded Answers



Accurate & Relevant



DEPARTMENT DOCUMENTS



Turning departmental documents into a secure, intelligent student support system.

Related Work



Common limitations in existing AI Chatbots and Assistants

Ref	Domain	AI/Model	API/Database	Key Limitation
1	Education + Healthcare	GPT-based LLM	OpenAI API	⚠️ Hallucination, low domain accuracy
2	Multi-domain	GPT, BERT, LLaMA	OpenAI / HuggingFace	⚠️ High computation, bias, cloud dependency
5	Education	ChatGPT	OpenAI API	⚠️ Unreliable for administrative decisions
6	Healthcare + Admin	GPT-3	OpenAI API (Cloud)	⚠️ (Cloud) Privacy concerns, cloud dependency
7	Education	NLP + LLM	Mixed APIs	⚠️ High system complexity
9	Domain-Specific Comm	GPT + Rule-based	OpenAI API	⚠️ Limited adaptation to new policies
15	Multi-domain	LLMs	OpenAI / HuggingFace	⚠️ Bias + Hallucination, cloud cost

Research Gap – This is What We Targeted to Develop

⚠ Research Gaps in Existing Systems

❌ Heavy reliance on cloud-based proprietary APIs

❌ High recurring costs

❌ Frequent hallucinations confident but incorrect answers

❌ Lack of strict domain restriction

❌ High computational overhead and complex maintenance

❌ Poor suitability for resource-limited



✅ Zero-cost & offline

✅ Strict domain gating

✅ Built with Ollama (Llama 3.2) + Sentence Transformers + FAISS

✅ Easy admin panel: can upload PDFs/DOCX

✅ Suitable for modest hardware

● Our Solution

● Gap



The Problem: Overburdened Departmental Query Handling



**High volume of
repetitive queries
every day.**

**Traditional
methods create
serious issues.**

**Heavy workload on
faculty and
administrative staff.**





Our Solution: DEPTORA



Department-specific

Designed exclusively for departmental needs.



Document-grounded

Answers are grounded in official departmental documents.



Domain restricted

Answers only department-related queries.



Privacy-first

Your data stays local.
No external sharing.



Local LLM

Powered by local LLMs running on your own infrastructure.



Automated support

Reduces repetitive workload and improves efficiency.

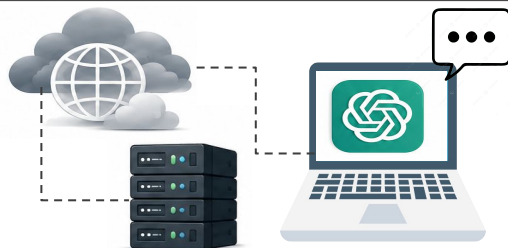
A locally hosted AI assistant that answers departmental queries using official institutional information.



Why Not Just Use Chatgpt?



GENERAL AI (Chatgpt)



When is the deadline of thesis submission?



Cloud dependent



May hallucinate



General Purpose



External data processing



No strict domain control



DEPTORA (Department Assistant)



Fully local



Grounded in documents



Department Specific



Local data



Domain Restriction



WHAT DID WE BUILD?

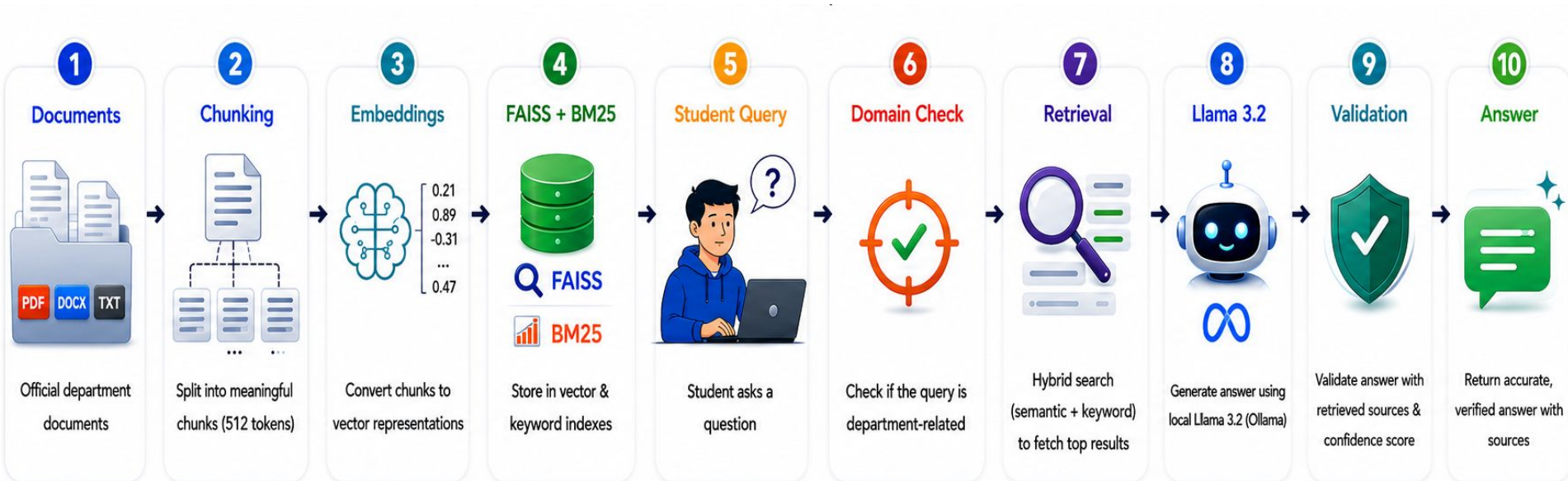


Our Goal:
Deliver accurate, reliable and secure departmental assistance automatically.





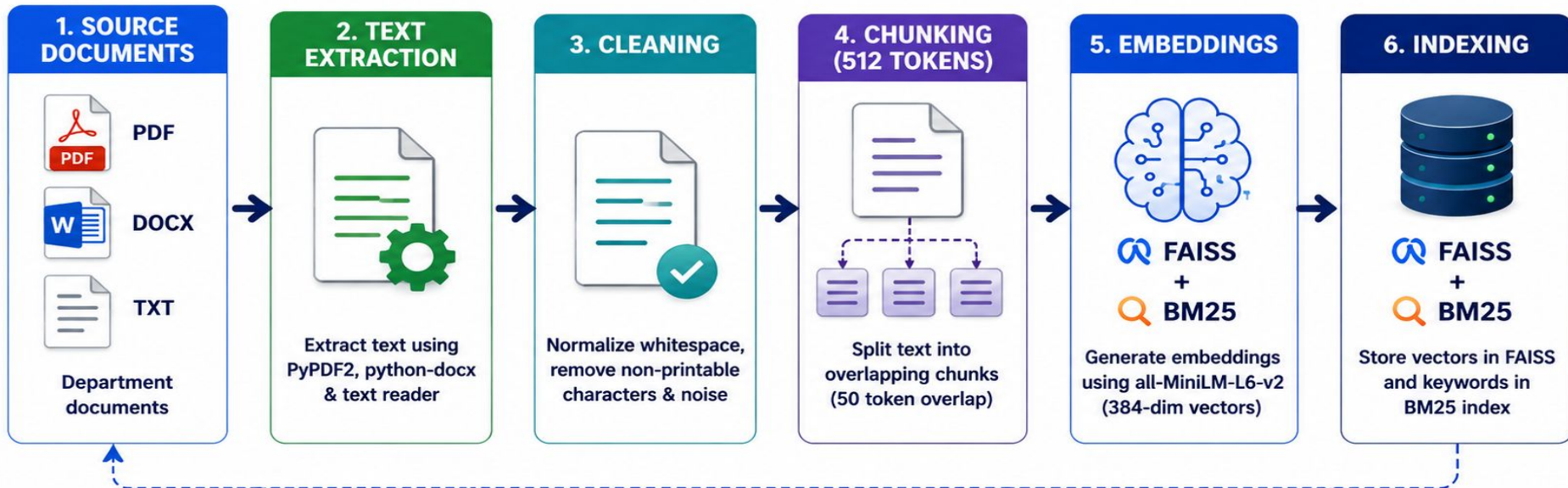
How DEPTORA Works





From Documents to Knowledge

We transform raw documents into searchable, structured knowledge.



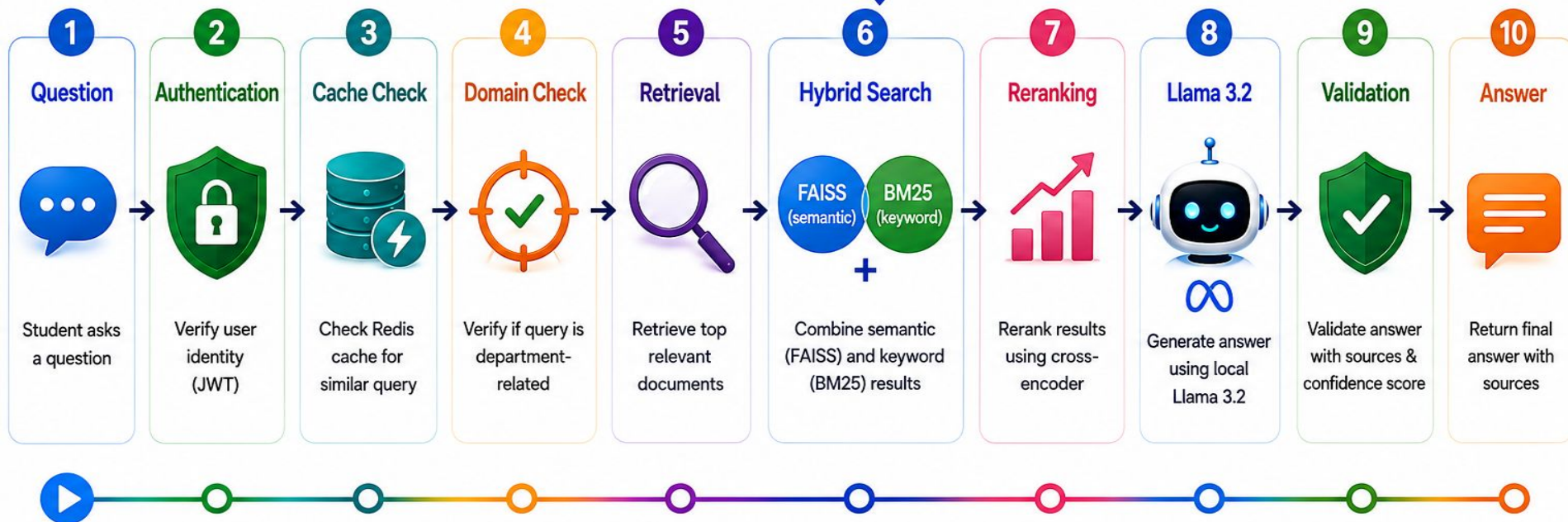
AUTOMATIC RE-INDEXING WHEN NEW DOCUMENTS ARE ADDED.
ALWAYS UP-TO-DATE KNOWLEDGE BASE!



What Happens When a Student Asks?



"What is the process for submitting my thesis?"





Why Hybrid Retrieval?

Combine the strengths of semantic understanding and exact matching to get the most relevant results.



FAISS

Semantic Search

Understands meaning

Example



"When can I meet my advisor?"



matches:
"Advising schedule"



Hybrid Score

60% Semantic + 40% Keyword



BM25

Keyword Search

Matches exact terms

Useful for:

- ✓ CSE407
- ✓ Form numbers
- ✓ Policy names
- ✓ Exact terminology



Cross-Encoder Reranking

Deep relevance understanding for final ranking



Final Top Results

Most relevant and accurate answers



DEPTORA Doesn't Answer Everything

Example-1



"What are the prerequisites for CSE407?"



Answer Provided



The prerequisite for CSE407 is completion of CSE305 and CSE306 with a minimum grade of C.

Sources

- Undergraduate Curriculum.pdf (p. 12)
- Course Flowchart.pdf (p. 5)
- Academic Policy.docx (p. 8)



DEPTORA



Hard Reject

- Block obvious out-of-domain queries



Keyword Check

- Match with allowed departmental terms



Semantic Check

- Understand meaning & intent



Context Check

- Verify against departmental context

Example-2



"What's the weather today?"



Query Rejected

I can only answer questions related to the department (academics, courses, policies, procedures, etc.).



Why Rejected?

The question is not related to any departmental topic.

How it works



Question
Received



DEPTORA
Checks



In Domain?
Score ≥ 0.35



ACCEPT
Answer

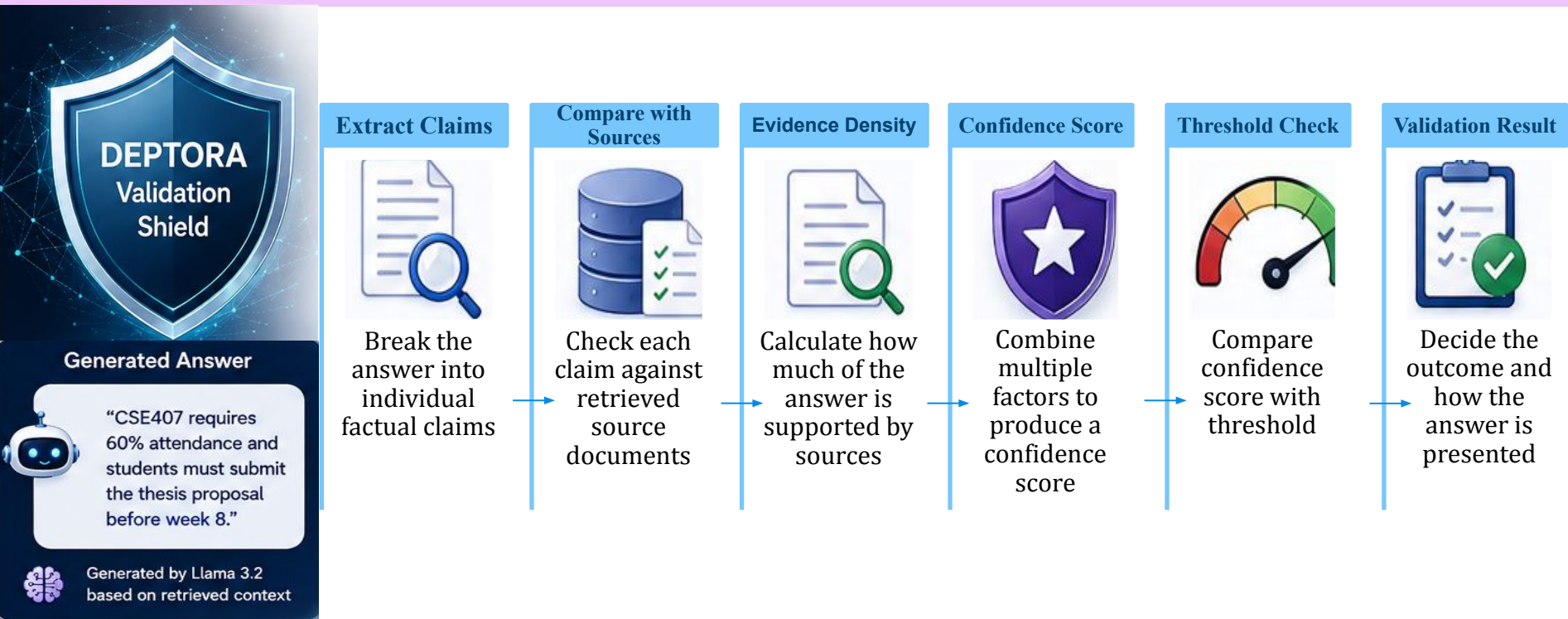
OR



REJECT
Don't Answer



HOW DO WE REDUCE HALLUCINATION?



We provide a source-based validation safeguard against unsupported claims.

We do not claim to eliminate hallucination completely.

We focus on **transparency, source tracking, and confidence scoring** to keep answers reliable and accountable.



Privacy-First Architecture



SECURE ACCESS

 **JWT Authentication**

 **bcrypt Passwords**

 **Student/Admin Roles**



LOCAL DOCUMENT

Official institutional documents
PDF, DOCX, Txt



FAISS (SEMANTIC SEARCH)

Vector store for high-dimensional embeddings



BM25 (KEYWORD SEARCH)

Fast keyword search for exact matching



LLAMA 3.2 (LOCAL LLM)

Locally hosted language model (ollama)



SQLITE DATABASE

Stores users, chats, logs, settings and more



100% LOCAL PROCESSING
All components run on own infrastructure



NO EXTERNAL LLM API
No data is sent to any external service or API.

Student Experience

 DEPTORA

Join your department's AI network

Create your account to access the department knowledge base, get instant answers, and stay informed.

- Department FAQs
Instant answers to common questions
- Procedure Guidance
Step-by-step academic processes
- Document Search
Semantic search over dept. documents

© 2025 Department AI Assistant

Create Account

Register with your student credentials

Full Name

Student ID

Email Address

Password

Confirm Password

☐ I agree to the department's data usage policy. AI conversations are logged for quality assurance.

Create Account →

Already have an account? [Sign in](#)

 DEPTORA

Student Login

Sign in with your department credentials

Demo Credentials

student@test.edu / student123

Email Address

Password

Sign in →

Don't have an account? [Create one](#)

Admin login →

Hello! I'm here to help with all your department questions — courses, procedures, deadlines, and more.

Your department assistant awaits

Sign in to access AI-powered answers for your academic journey. All data is processed locally and securely.

- Instant answers to FAQs
- Document-grounded responses
- Available 24/7

© 2025 Department AI Assistant

 DEPTORA
RAE • Ollama

Department Assistant

RAE • Ollama

New Chat

QUICK QUESTIONS

- When are tuition fees due?
- What is the attendance policy?
- What is the minimum CGPA requirement to graduate?
- How do I get my official transcript?
- When is the advising?
- How do I apply for a scholarship?
- Who is the department head?

RESOURCES

- Procedures
- Documents

Test Student
ST-2024-1001

can you tell me the title of the course CSE106?

Unfortunately, I don't have this information in my knowledge base. Please check with the Computer Science & Engineering department's website or contact their office for the most accurate and up-to-date information about course titles and details.

Is there anything else I can help with?

Note: This answer has limited source backing. For critical information, please verify with the department office.

stt8d0f7099747b30baec0e4f5f1e0b0.pdf

what is the weather today?

Out-of-Domain Query Detected

I can only answer questions related to the department — courses, procedures, faculty, notices, deadlines, and academic policies.

Please rephrase your question to be department-related, or contact the main university helpdesk for general queries.

Ask about courses, procedures, deadlines...

Answers grounded in department documents via RAG - Powered by Ollama (local LLM)

 DEPTORA
RAE • Ollama

Department Assistant

RAE • Ollama

New Chat

QUICK QUESTIONS

- When are tuition fees due?
- What is the attendance policy?
- What is the minimum CGPA requirement to graduate?
- How do I get my official transcript?
- When is the advising?
- How do I apply for a scholarship?
- Who is the department head?

RESOURCES

- Procedures
- Documents

Test Student
ST-2024-1001

What is the attendance policy?

Q: What is the attendance policy?
A: Minimum 75% attendance is required to sit for final exams. Below 75% results in automatic NC (Not Complete) grade. Medical certificates must be submitted within 3 working days of the absence.

(Note: The provided information does not explicitly state the deadline for submitting medical certificates, but it mentions that they should be submitted 'within 3 working days' after an absence. Therefore, I assume that the deadline is 3 working days from the date of absence.)

I don't have this information in my knowledge base. Please contact [department office].

f09055e412c4438590c0bf50e9d25f6b0.pdf

789f02f800a48d30973f3c35e000b.pdf

7c0f0e3930b049f0c09ee289cb430b8d.pdf

09bb4730c0e0441c709d9f3b32008d93.pdf

What is the minimum CGPA requirement to graduate?

Based on available information, students must maintain a minimum CGPA of 2.5 to be eligible for graduation. Students with CGPA below 2.0 will remain academic probation status.

Ask about courses, procedures, deadlines...

Answers grounded in department documents via RAG - Powered by Ollama (local LLM)

Administrator Dashboard



Admin Portal

Restricted access — authorized personnel only

Demo Admin Credentials
admin@dept.edu / admin123

Admin Email

admin@dept.edu

Password

[Access Dashboard →](#)

All admin actions are logged and audited

Student? Sign in here

DEPTORA
Admin

NAVIGATION

- Dashboard
- FAQ Manager
- Procedures
- Documents
- Query Logs

Department Admin...
Administrator

Query Logs

Monitor all student interactions

244
Total Queries

3
Resolved (page)

0
Out-of-Domain (page)

17562ms
Avg Response

Q Search queries... ALL Resolved Out-of-Domain Partial

QUERY	USER	STATUS	METHOD	SOURCES	TIME	DATE
who is the dean of cse department?	2	Resolved	-	3 src	62612ms	07:19 AM
are there any public holidays in th...	2	Resolved	-	3 src	135567ms	07:04 AM
when is the advising?	2	Resolved	-	4 src	6ms	06:55 AM
What is the minimum CGPA requir...	2	Resolved	-	4 src	62ms	06:56 AM
when is the advising of summer 2...	2	Resolved	-	4 src	152375ms	05:49 AM
when is the advising?	2	Resolved	-	4 src	28ms	05:36 AM
How do i register for thesis?	2	Resolved	-	4 src	4ms	05:12 AM
when is the advising?	2	Resolved	-	4 src	9ms	05:12 AM
when is the advising of summer 2...	2	Resolved	-	4 src	5ms	05:08 AM
How do i register for thesis?	2	Resolved	-	4 src	7ms	12:29 PM
When are tuition fees due?	2	Resolved	-	3 src	13ms	12:29 PM
Can i take courses from other dep...	2	Resolved	-	4 src	15ms	12:29 PM

DEPTORA
Admin

NAVIGATION

- Dashboard
- FAQ Manager
- Procedures
- Documents
- Query Logs

Department Admin...
Administrator

Dashboard

System overview

244
Total Queries (30d)

34
Resolved Queries
93.8% resolution rate

0
Out-of-Domain Rejected

1
Active Students

8
Active FAQs

5
Indexed Documents

Recent Activity

- who is the dean of cse department?
resolved - 62612ms
07:19 AM
- are there any public holidays in the academic calendar?
resolved - 135567ms
07:04 AM
- when is the advising?
resolved_cached - 6ms
06:55 AM
- What is the minimum CGPA requirement to graduate?
resolved_cached - 62ms
06:56 AM
- when is the advising of summer 2028?
resolved - 152375ms
05:49 AM
- when is the advising?
resolved_cached - 28ms
05:36 AM

System Performance

eval: 8/7/2026

Domain Restriction Accuracy (labeled text set) 94.5%

Document Extraction Success Rate 100%

Cosine Hit Rate (243 lookups) 89.5%

Avg Response Validation Confidence 56%

14245.4ms avg response (p95: 94169ms, n=244)

Quick Actions

- Add FAQ
- Upload Doc
- View Logs
- Procedures

DEPTORA
Admin

NAVIGATION

- Dashboard
- FAQ Manager
- Procedures
- Documents
- Query Logs

Department Admin...
Administrator

Document Upload

Manage the FAQ knowledge base

5
Total

5
Indexed

0
Processing

765
Total Chunks

Drag & drop files here
PDF, DOCX, TXT - Max 10MB per file

[Click to browse](#)

Pipeline: Upload → Extract Text → Chunk (512 tokens) → SentenceTransformers → FAISS Index

Knowledge Base Documents

DOCUMENT	STATUS	CHUNKS	SIZE	UPLOADED	ACTIONS
ewi_undergrad_rate_bulletin-14th-edition.pdf 524 pages	Indexed	759	5.8 MB	-	View Edit
Final exam Schedule.pdf 1 pages	Indexed	1	84 KB	-	View Edit
academic calendar.pdf 1 pages	Indexed	2	133 KB	-	View Edit
te circular.pdf 1 pages	Indexed	1	148 KB	-	View Edit
ra circular.pdf	Indexed	2	151 KB	-	View Edit



How Did We Evaluate DEPTORA?



5

Documents
official departmental
documents ingested
(PDF,DOCX,TXT)



35

Domain Test Queries
Carefully crafted in-domain and
out-of-domain queries for
domain restriction evaluation



503

Logged Queries
Real queries from students
during testing and pilot
deployment



5

FAQ Comparisons
Manual FAQ pairs used for
answer accuracy and
hallucination validation



CPU-only Ollama

Local LLM inference using
Ollama (Llama 3.2)
No GPU required



Docker Compose

All services containerized
for easy deployment and
reproducibility



Windows Host

Experiments conducted on
Windows environment
Intel i5, 16GB RAM



Local Deployment

Fully local, no external APIs
or cloud services used
during evaluation



Goals

Test accuracy,
domain restriction,
hallucination rate,
and response time



Metrics

Answer Accuracy
Domain Accuracy
Hallucination Rate
Response Time



Method

Manual evaluation
Automated logging
Source verification



Testing Period
2 Weeks

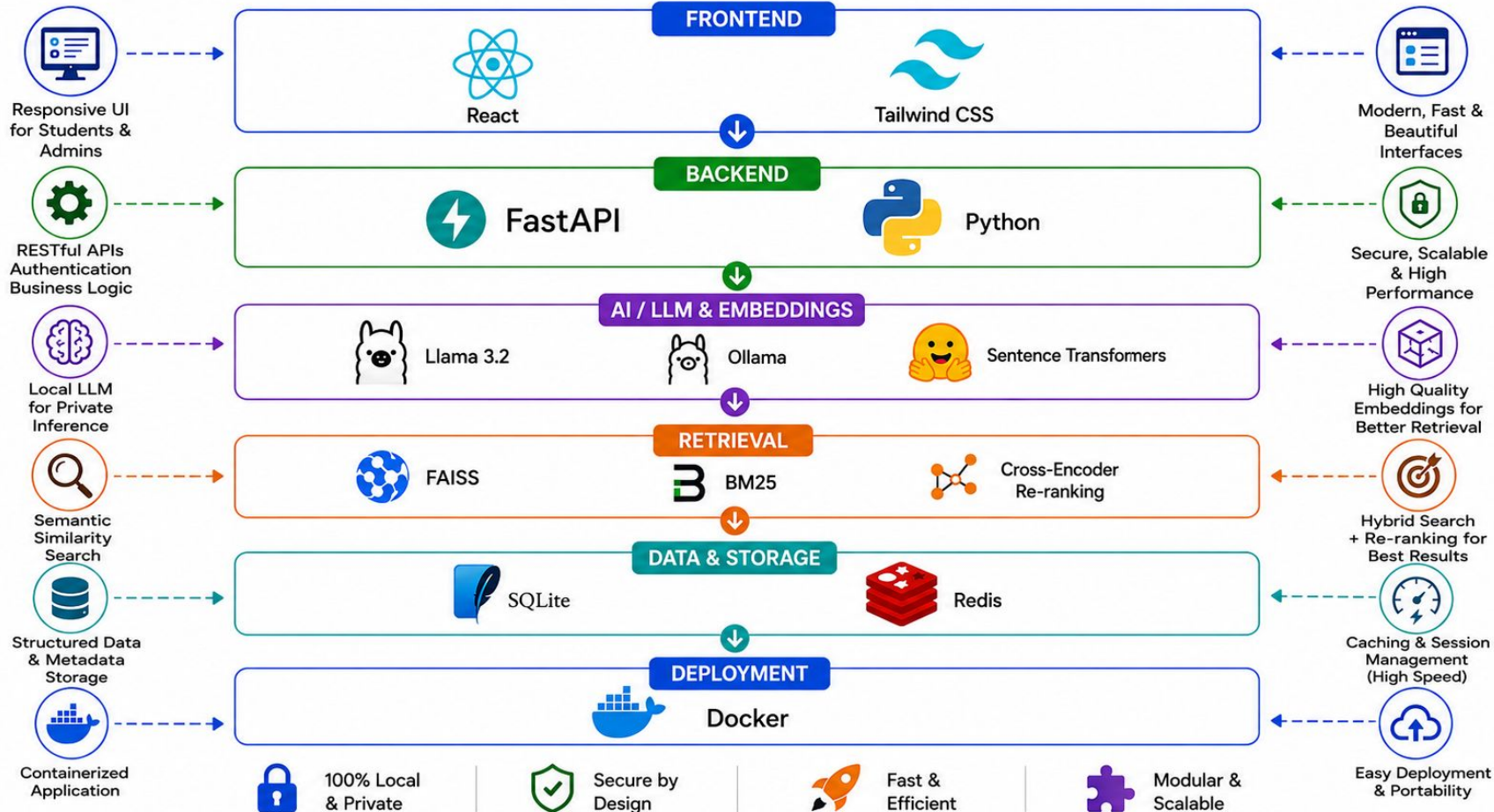


**Total
Evaluations**

150+ evaluations
across all metrics
and scenarios



Technology Behind DEPTORA





What Did We Achieve?

System Performance

eval: 8/11/2026

Domain Restriction Accuracy (labeled test set)

94.3%



Document Extraction Success Rate

100%



Cache Hit Rate (508 lookups)

92.3%



Avg Response Validation Confidence

56%



● 9040.3ms avg response (p95: 76791ms, n=503)

**This is a screenshot of the evaluation tab from the admin dashboard.
All the results are measured in the system.**





What Did We Achieve?



94.3%

Domain Accuracy

Excellent domain restriction ensures DEPTORA answers only in-domain questions.



92.3%

Cache Hit Rate

High cache efficiency significantly reduces response time and load.



0.58

Validation Confidence

Source-based validation confidence for generated answers (*higher is better*).



46.8

BLEU

Strong n-gram overlap indicates high similarity with reference answers



59.6

ROUGE-L

High recall-oriented score shows better coverage of important information



Accurate

Answers the right questions with high domain accuracy.



Efficient

Smart caching delivers fast and responsive system.



Reliable

Validation layer adds trust and reduces hallucinations.



High Quality

Strong similarity with reference answers (BLEU & ROUGE-L).



Overall Impact

A robust, trustworthy and domain-focused AI assistant.



DEPTORA delivers **accurate**, **reliable** and **efficient** answers while strictly staying in-domain





Limitations



 **DATA**



Small
5-document corpus



 **RETRIEVAL**



No completed
labeled retrieval
dataset



 **EVALUATION**



Human answer grading
& hallucination
evaluation pending



 **SYSTEM**



Single-server
+ CPU-only
+ no OCR



DEPTORA is a **working prototype** -not yet production-scale



FUTURE WORK



Conclusion

94.3%

Domain
Accuracy

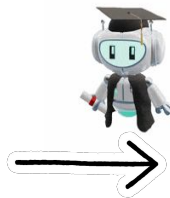
92.3%

Cache Hit
Rate

9.04s

Average
Latency

Before



After



DEPTORA provides an efficient, accurate, and maintainable solution that reduces administrative workload while maintaining institutional data sovereignty and reliability.



Thank You!!

DeptAI — Department Assistant

localhost:3000

Chat

 **DEPTORA**

FeaturesHow it worksAboutAdminSign inGet Started →

AI-Powered • Available 24/7 • No queue

Your department's smartest secretary

Get instant answers to any university question — transcripts, registration, leave applications, schedules, and more. No waiting. No frustration.

Start Chatting →I have an account

Trusted by the Department of Computer Science & Engineering

< 15 Response time	24/7 Availability	95% Match rate	100+ FAQs covered
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Built for Academic Departments

87°F
Mostly cloudy

Search



1:19 PM
8/13/2026

65%

Any Questions?



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